Customer Churn Prediction Interim Report

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1. Understanding the Business Problem

1. Defining the Problem Statement

AlphaCom is currently facing a significant increase in customer churn, which is negatively impacting both revenue and brand loyalty. Despite offering competitive services, the company is unable to effectively retain customers due to the lack of actionable insights into churn behavior.

Customer churn is not driven by a single factor but rather a complex interaction of multiple variables such as:

- Service usage patterns
- Billing preferences
- Contract types
- Customer demographics

The absence of a predictive and data-driven approach has resulted in reactive decision making, where retention strategies are implemented only after customers have already churned.

Problem Statement:

To identify customers who are likely to churn and understand the key factors influencing their decision, enabling proactive and targeted retention strategies.

2. Need of the Study / Project

This project is essential for transforming AlphaCom's approach from reactive to proactive customer management.

Key Needs:

- Early Identification of At-Risk Customers
- Data-Driven Decision Making
- Optimization of Retention Strategies
- Revenue Protection
- Customer Experience Enhancement

Without this study, AlphaCom risks:

- Continued revenue decline
- Increased customer acquisition costs

3. Understanding Business & Social Opportunities

Business Opportunities

- Targeted Marketing Campaigns
- Improved Customer Lifetime Value (CLV)
- Product & Service Optimization
- Competitive Advantage

Social Opportunities

- Better Customer Satisfaction

2. Exploratory Data Analysis

1. Data Collection and Background

- **Context:** AlphaCom, a telecommunications provider, is facing a significant increase in customer churn. This trend negatively impacts revenue and brand reputation in a highly competitive market.
- **Data Source:** The analysis is based on the customer_churn.csv dataset, which contains comprehensive information on customer demographics, service usage, billing, and contracts.

2. Data Overview

- **Dataset Dimensions**: The dataset consists of 12,055 rows and 20 columns.
- **Demographics**: Gender, SeniorCitizen, Partner, Dependents.
- **Services**: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Account Information**: Tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges.
- **Target Variable**: Churn

3. Univariate Analysis

1. Distribution of 'Churn'

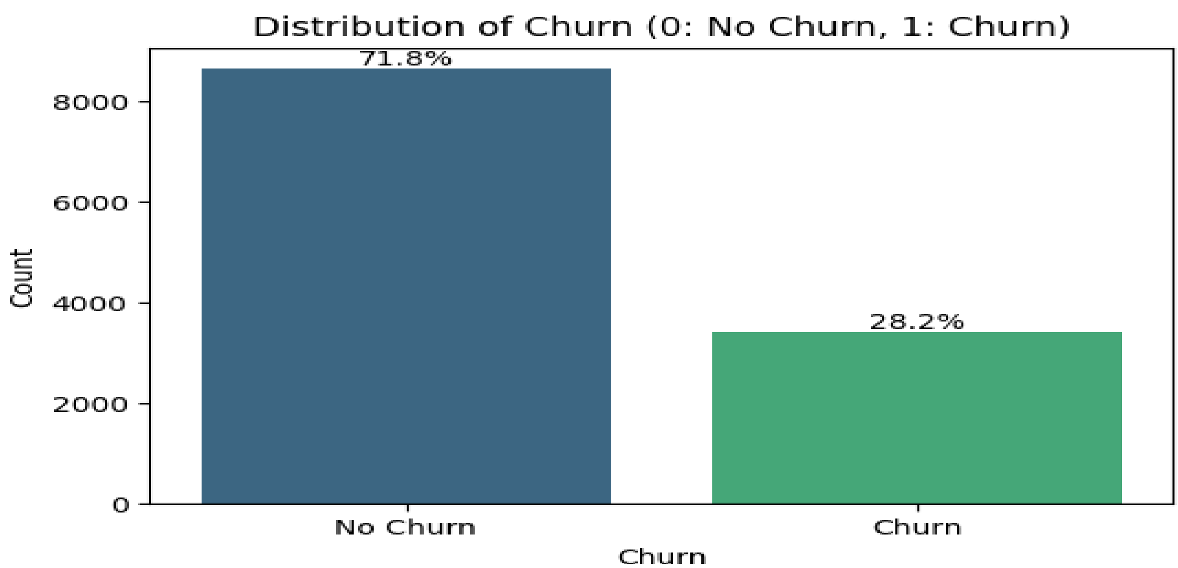


Fig. 1: Distribution of 'Churn'

- Approximately 28% of the customers have churned (represented by '1'), while about 72% have not churned (represented by '0').

2. Distribution of 'SeniorCitizen'

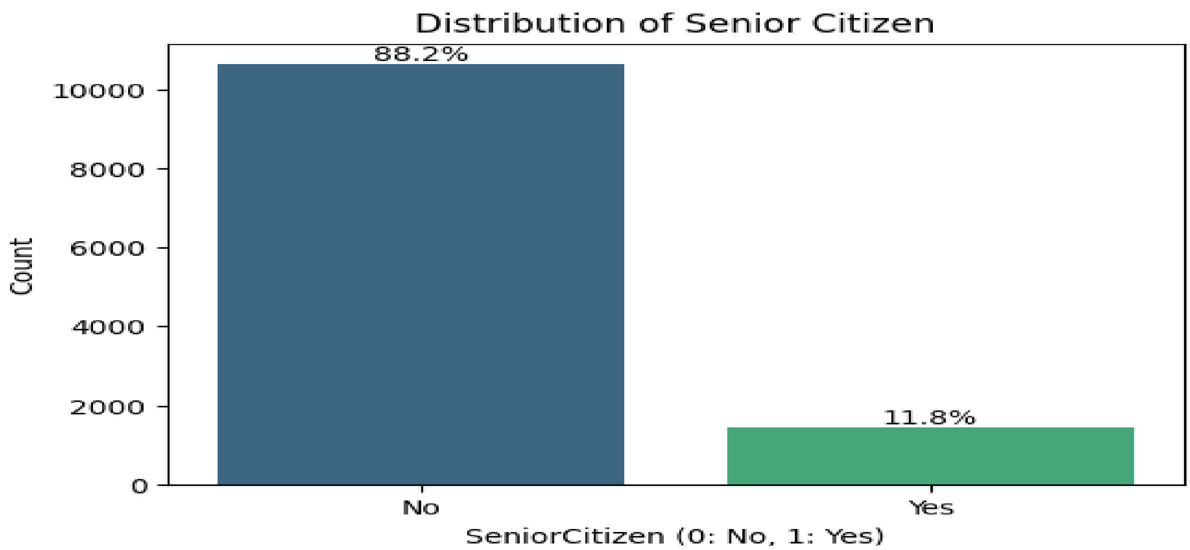


Fig. 2: Distribution of 'SeniorCitizen'

- The plot indicates that the vast majority of customers (88.0%) are not senior citizens, while only a small percentage (12.0%) are senior citizen.

3. Distribution of 'tenure'

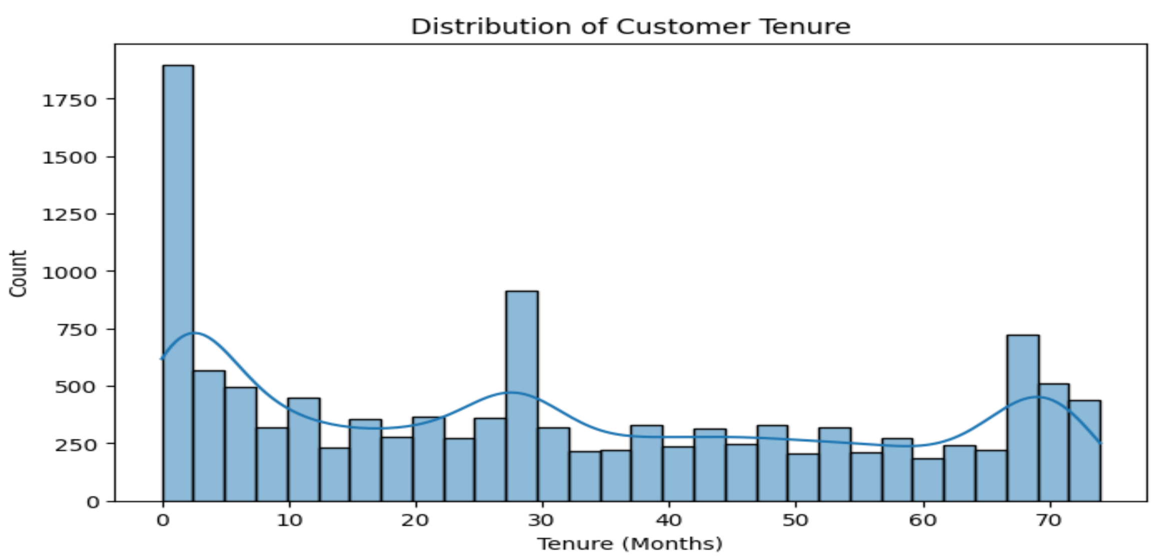


Fig. 3: Distribution of 'tenure'

- The distribution of 'tenure' shows a bimodal shape, with a significant number of customers having very short tenure (around 0-5 months) and another peak for customers with longer tenure (around 65-70 months). This suggests that many customers either churn early or stay for a long time.

4. Distribution of 'Contract'

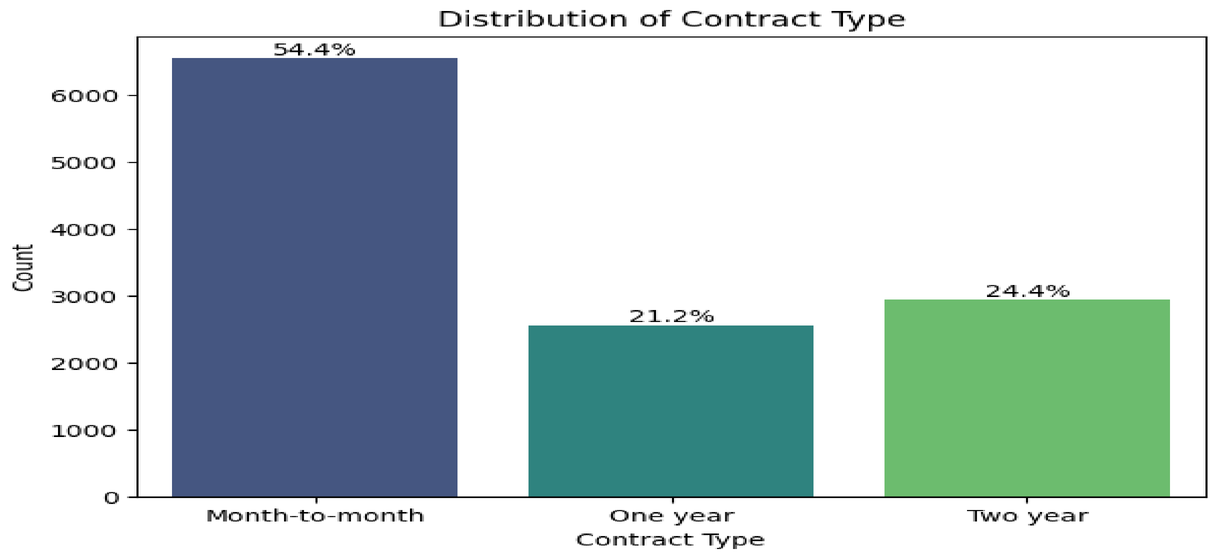


Fig. 4: Distribution of 'Contract'

- The distribution of 'Contract' types shows that 'Month-to-month' contracts are the most prevalent, followed by 'Two year' and then 'One year' contracts. This indicates that a significant portion of customers prefer shorter, more flexible contract durations.

5. Distribution of 'PaymentMethod'

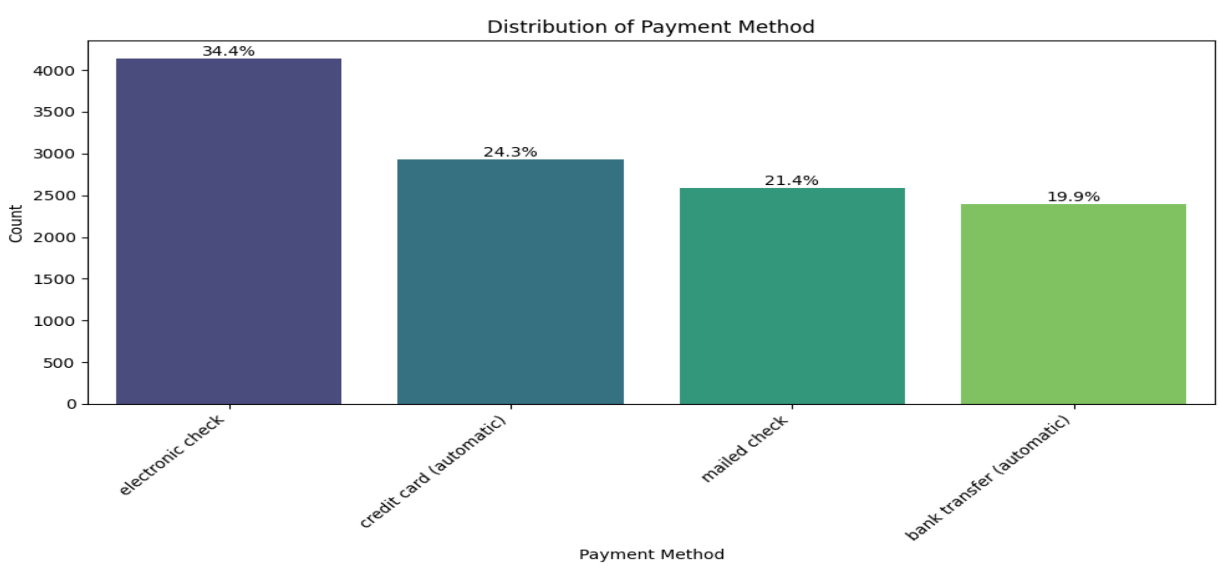


Fig. 5: Distribution of 'PaymentMethod'

- The 'electronic check' is the most common payment method, followed by 'mailed check', 'bank transfer (automatic)', and 'credit card (automatic)'.

6. Distribution of 'InternetService'

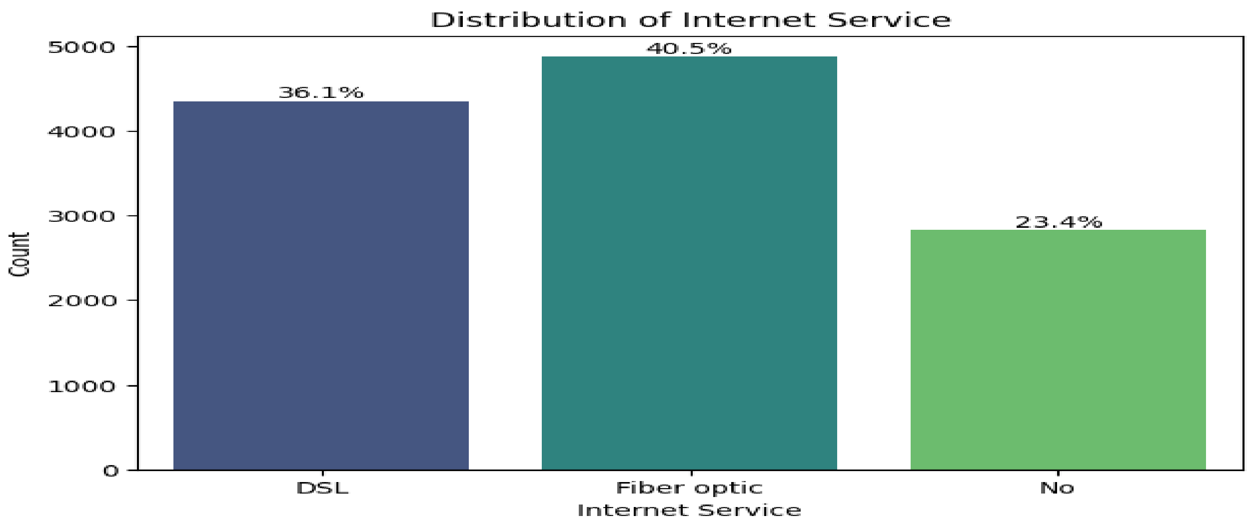


Fig. 6: Distribution of 'InternetService'

- The 'Fiber optic' service has the highest number of customers, followed by 'DSL', and then a significant portion of customers with 'No' internet service.

7. Distribution of 'MonthlyCharges'

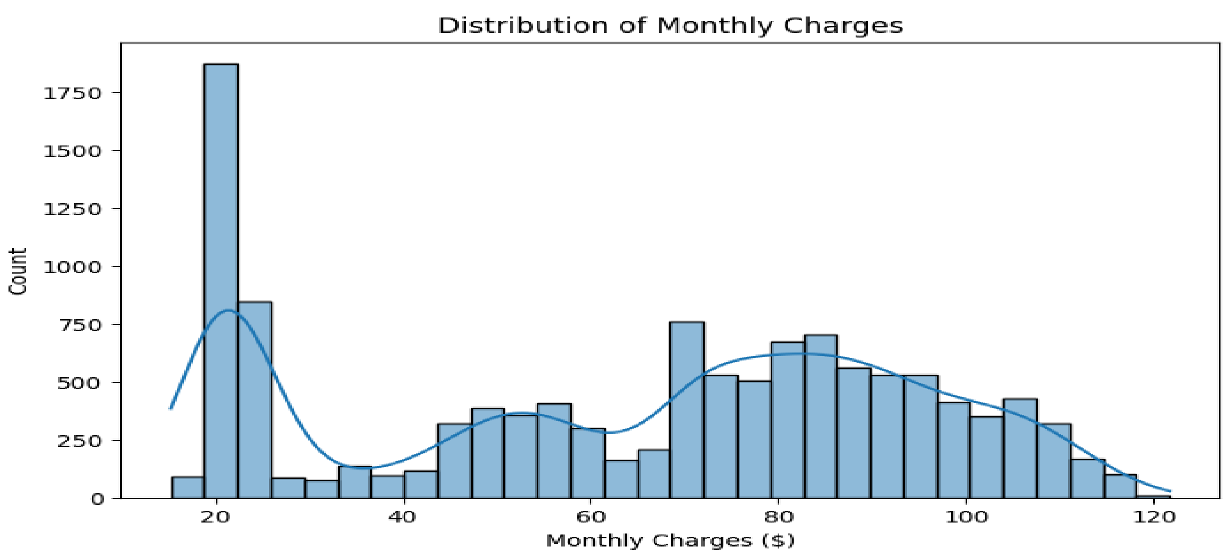


Fig. 7: Distribution of 'MonthlyCharges'

- we can observe that there's a bimodal distribution. There's a significant peak at lower monthly charges (around 20 Dollar) and another broader peak ranging from 70 Dollar to 100 Dollar. This suggests that customers tend to either opt for very basic, low-cost plans or more comprehensive, higher-cost packages. Very few customers fall into the mid-range of monthly charges.

8. Distribution of 'TotalCharges'

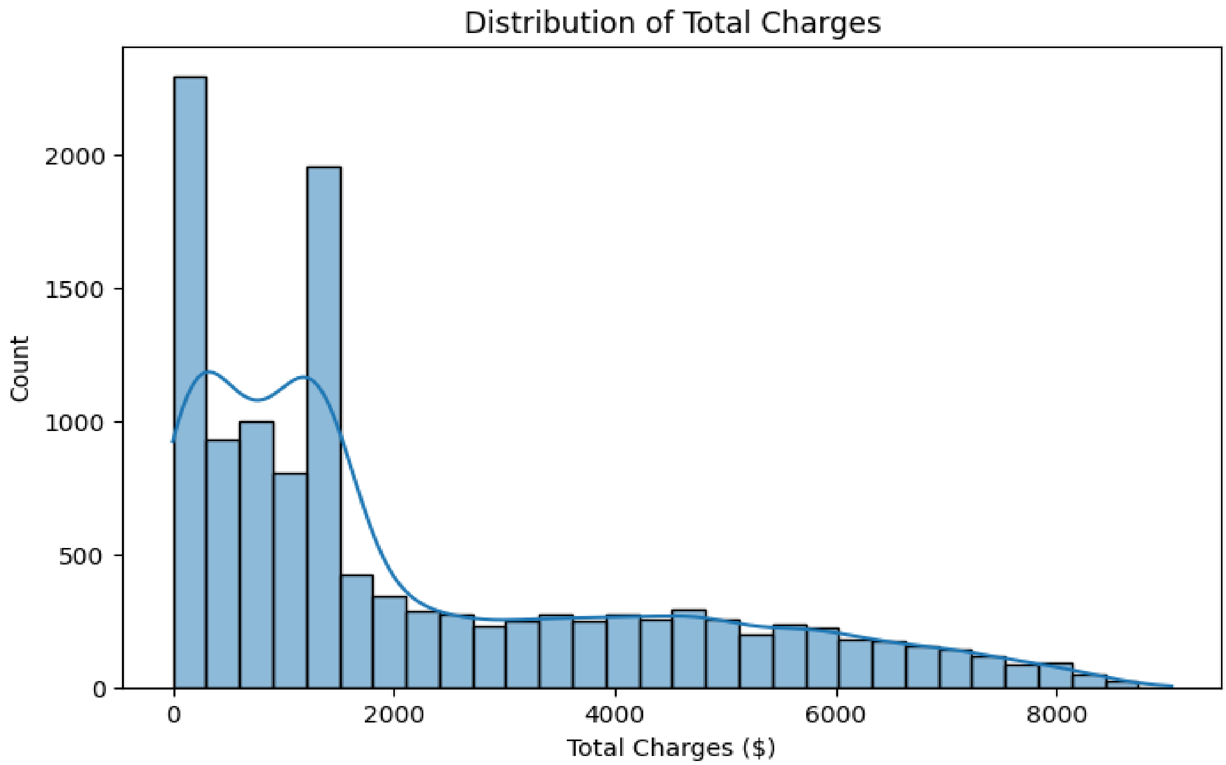


Fig. 8: Distribution of 'TotalCharges'

- we can see a right-skewed distribution, indicating that a large number of customers have lower total charges, while fewer customers accumulate very high total charges.

4. Bivariate Analysis

1. Churn by Contract

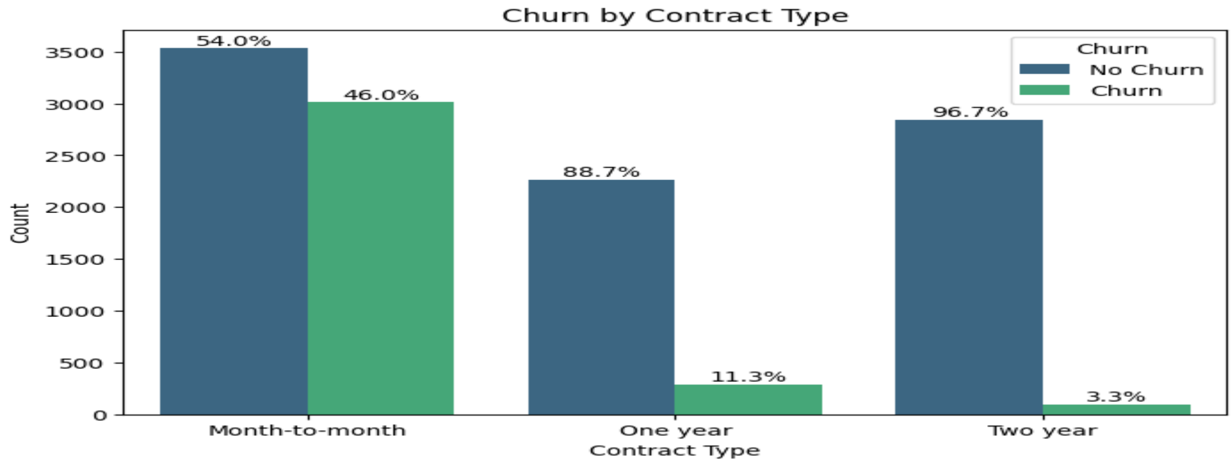


Fig 9: Churn by Contract

- Month-to-month contracts have the highest churn rate, with approximately 46.0%.
- One year contracts show a much lower churn rate of about 11.3%.
- Two year contracts have the lowest churn rate at approximately 3.3%.

This analysis highlights that contract duration is a critical factor influencing customer churn, with longer contracts leading to higher retention.

2. Churn by Tenure

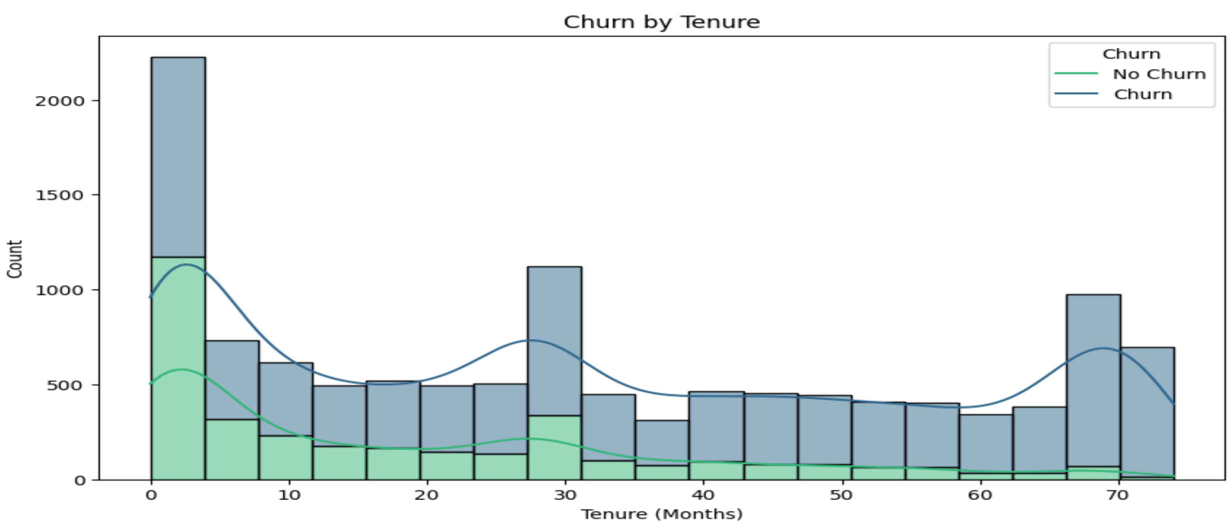


Fig 10: Churn by Tenure

- The histogram shows that a significant portion of churn occurs among customers with very short tenures (0-5 months). As tenure increases, the proportion of churned customers generally decreases, and customers with longer tenures (60-70 months) show a much lower churn rate. This indicates that newer customers are more prone to churn, while long-term customers are more loyal.

3. Churn by InternetService

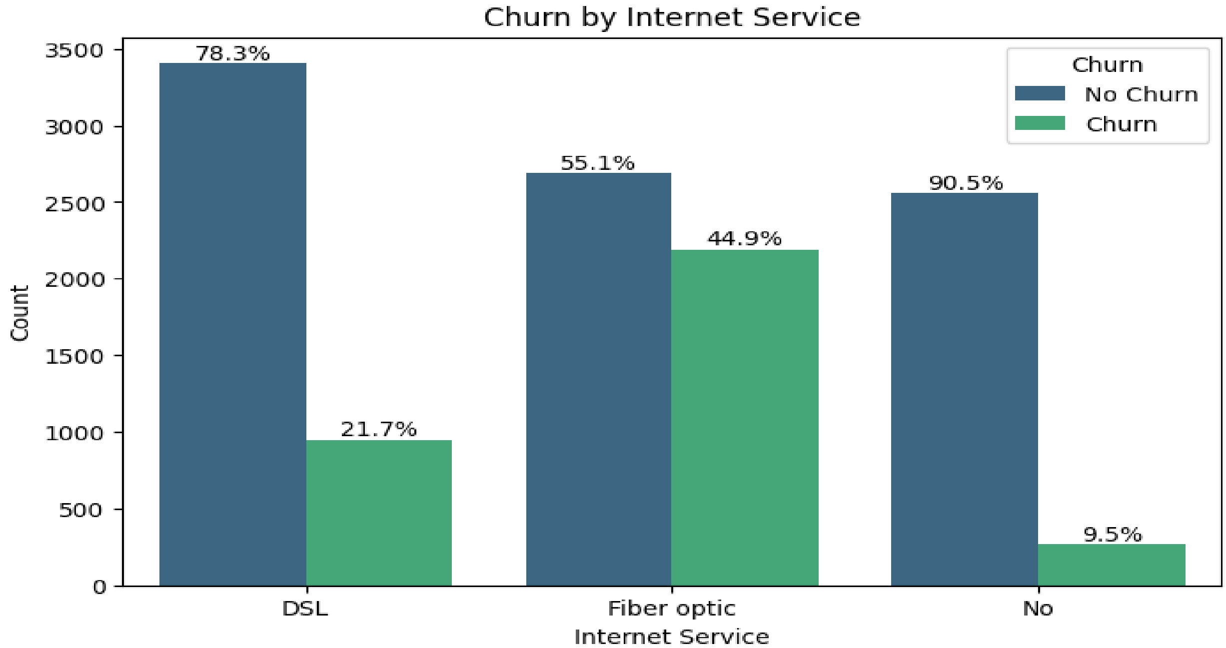


Fig 11: Churn by InternetService

- Fiber optic internet service has the highest churn rate at approximately 44.9%. This indicates that almost half of the customers using fiber optic churn.
- DSL internet service has a significantly lower churn rate of about 21.7%.
- Customers with No internet service have the lowest churn rate at approximately 9.5%.

This analysis clearly shows that customers with 'Fiber optic' internet service are far more likely to churn than those with 'DSL' or no internet service.

4. Churn by TechSupport

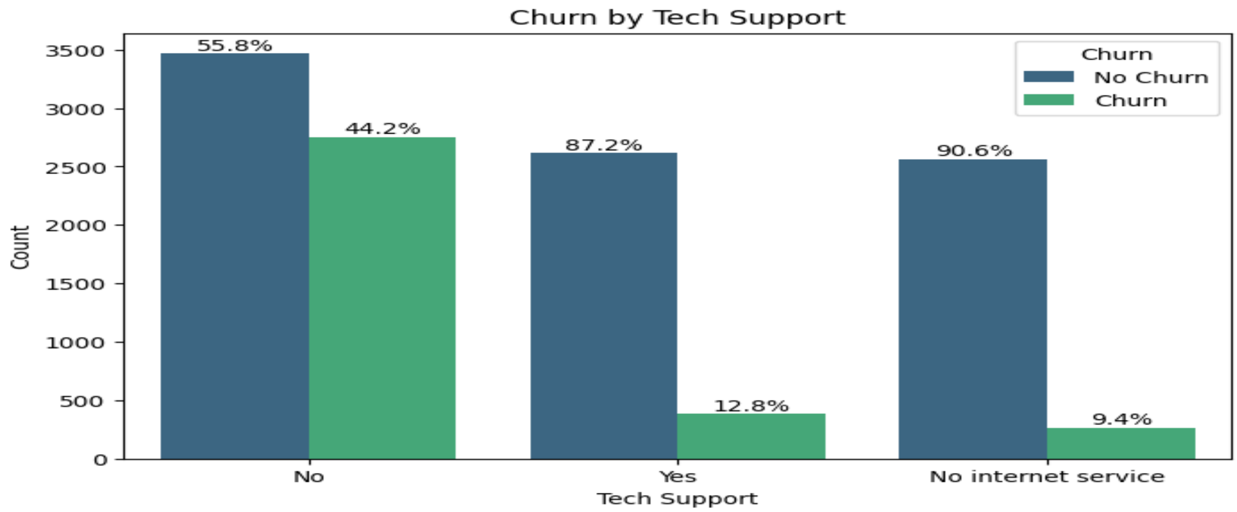


Fig 12: Churn by TechSupport

- Customers with 'No Tech Support' have a significantly higher churn rate of approximately 44.2%.
- Customers with 'Yes Tech Support' have a much lower churn rate of about 12.8%.
- Customers with 'No internet service' have the lowest churn rate at approximately 9.4%.

This highlights the importance of technical support in customer retention, as customers without it are considerably more likely to churn.

5. Churn by OnlineSecurity

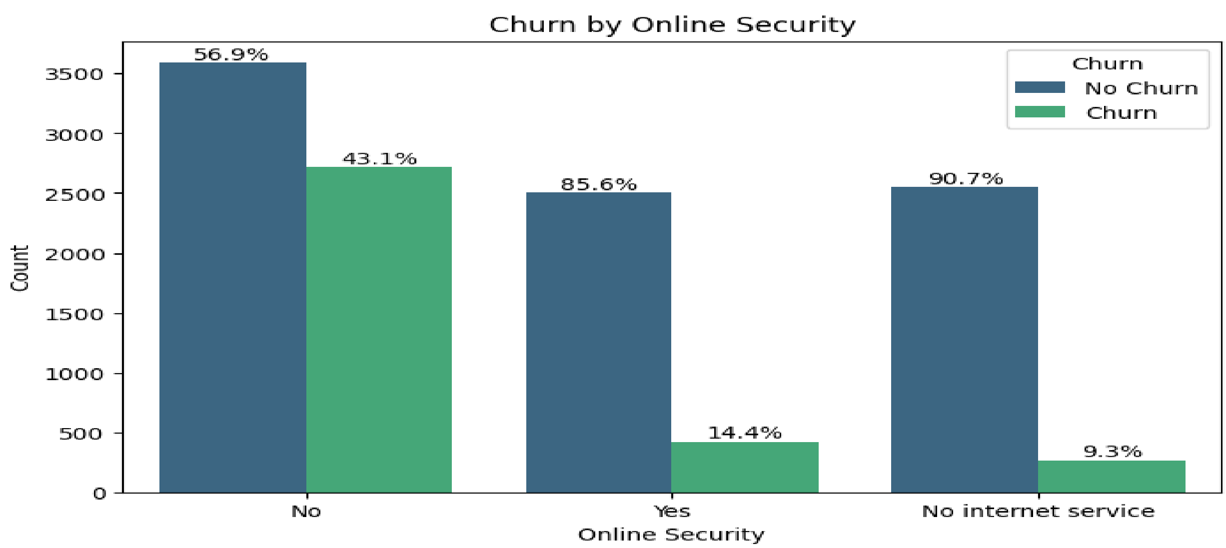


Fig 13: Churn by OnlineSecurity

- Customers with 'No Online Security' have a significantly higher churn rate of approximately 43.1%.
- Customers with 'Yes Online Security' have a much lower churn rate of about 14.4%.
- Customers with 'No internet service' have the lowest churn rate at approximately 9.3%.

This indicates that the presence of online security is a strong factor in customer retention, with customers lacking this service being more prone to churn. Customers who do not have internet service at all are the least likely to churn.

6. Churn by PaymentMethod

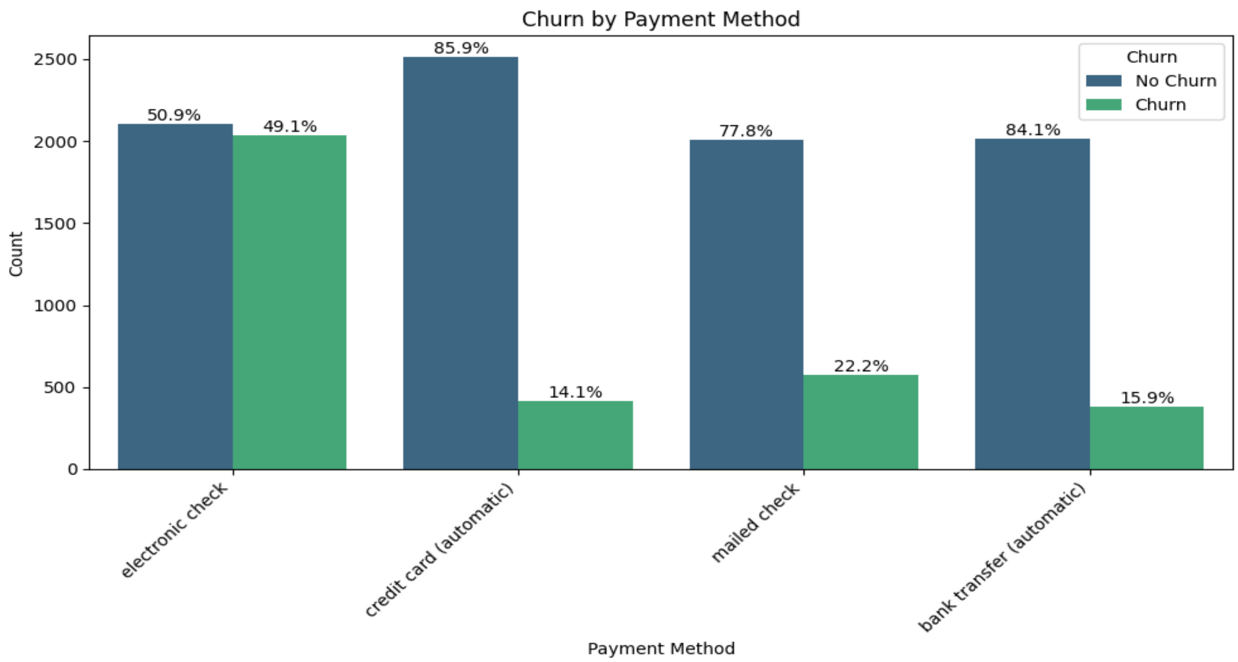


Fig 14: Churn by PaymentMethod

- Electronic check is associated with the highest churn rate, approximately 49.1%.
- Mailed check has a churn rate of about 22.2%.
- Bank transfer (automatic) has a churn rate of about 15.9%.
- Credit card (automatic) has the lowest churn rate, approximately 14.1%.

This analysis suggests that manual payment methods, especially electronic checks, are strongly correlated with higher churn, while automatic payment methods like bank transfers and credit cards are associated with greater customer retention.

7. Churn by MonthlyCharges

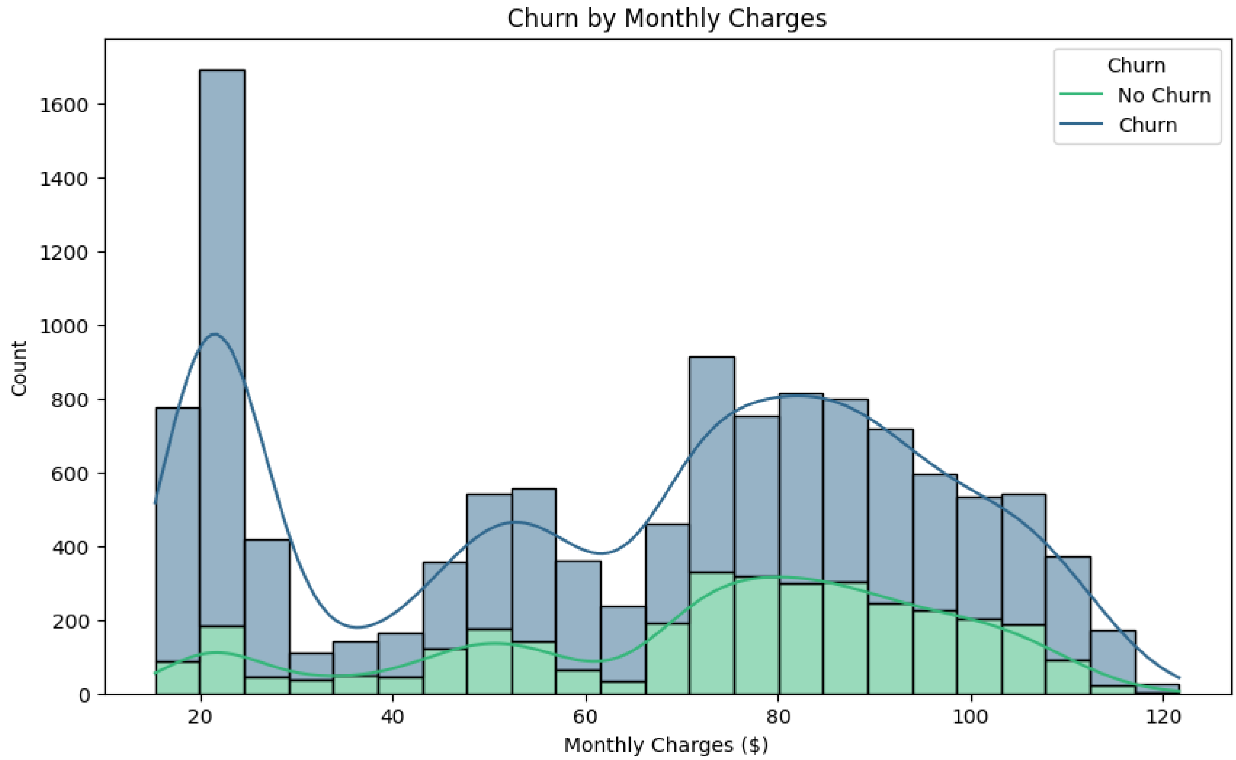


Fig 15: Churn by MonthlyCharges

- The plot shows that customers with lower monthly charges have a higher propensity to churn. As monthly charges increase, the proportion of churned customers decreases, indicating that customers with higher spending are more loyal.

5. Multivariate Analysis

1. Heatmap of Numerical Features

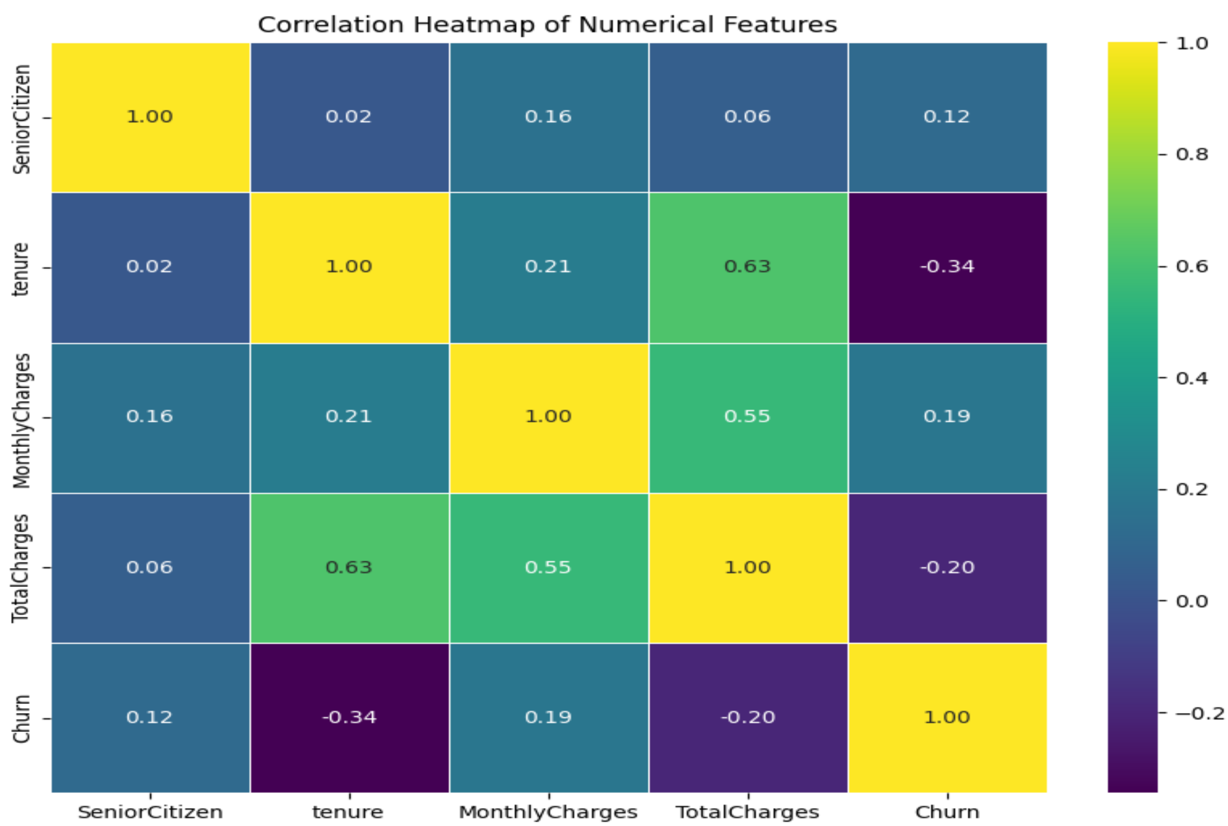


Fig 16: Heatmap of Numerical Features

From the correlation heatmap, we can observe the following:

- **Tenure and TotalCharges**: There is a strong positive correlation (0.63) between 'tenure' and 'TotalCharges'. This is expected, as customers with longer tenure are likely to accumulate higher total charges.
- **MonthlyCharges and TotalCharges**: There is a strong positive correlation (0.55) between 'MonthlyCharges' and 'TotalCharges'. This also makes sense, as higher monthly charges contribute to higher total charges over time.
- **MonthlyCharges and Churn**: There is a moderate positive correlation (0.19) between 'MonthlyCharges' and 'Churn'. This suggests that customers with higher monthly charges are slightly more likely to churn, which aligns with the bivariate analysis insights.
- **Tenure and Churn**: There is a moderate negative correlation (-0.34) between 'tenure' and 'Churn'. This indicates that customers with longer tenure are less likely to churn.

- **SeniorCitizen and Churn**: There is a weak positive correlation (0.12) between 'SeniorCitizen' and 'Churn'. This suggests that senior citizens are slightly more likely to churn, consistent with the earlier bivariate analysis.

Other correlations among numerical features are relatively weak, indicating that these variables largely have independent effects on churn or other numerical variables.

2. Pair Plot of Numerical Features

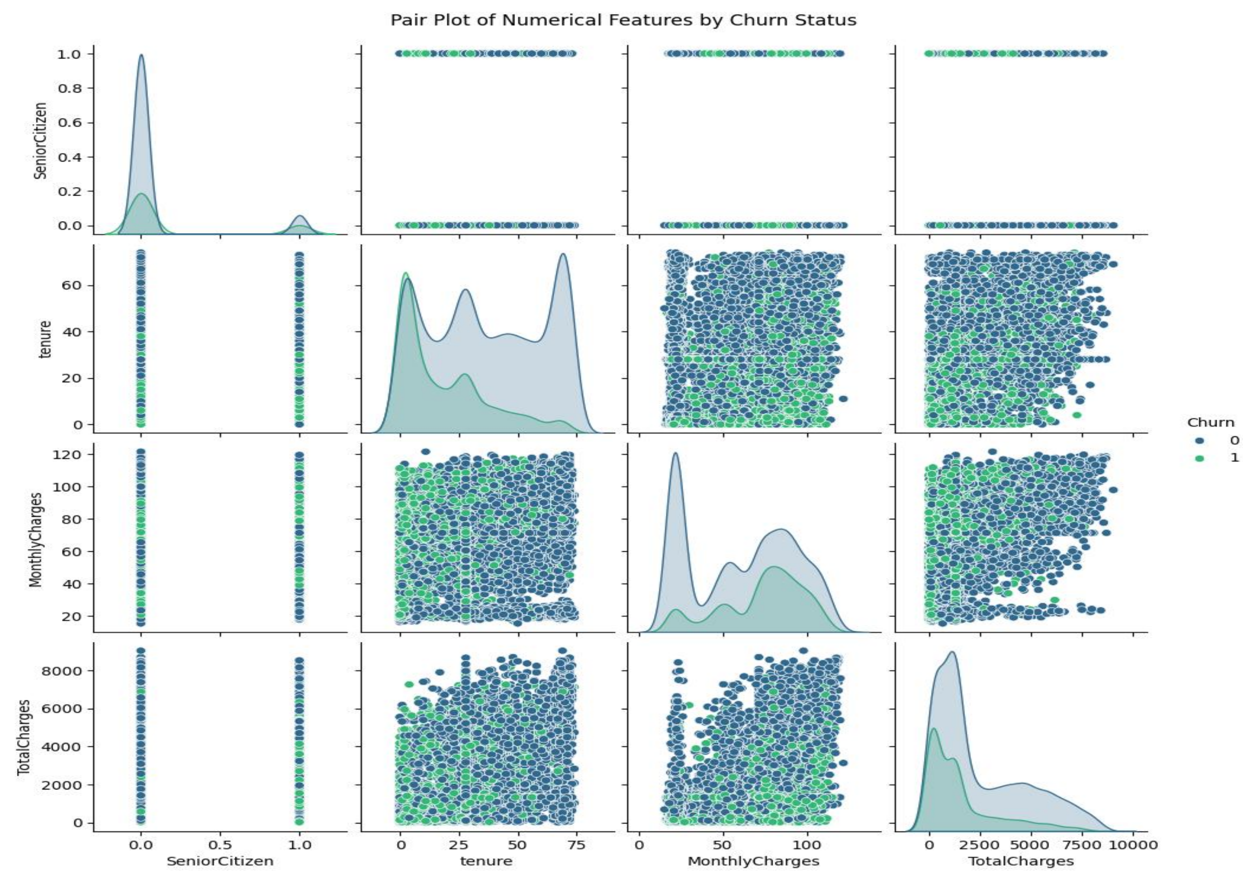


Fig 17: Pair Plot of Numerical Features

This pair plot provides a comprehensive view of the relationships between all numerical features ('SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges') and their distributions, with the 'Churn' status highlighted. Key observations include:

- **Diagonal plots**: Show the distribution of each numerical feature for churned and non-churned customers. For instance, 'tenure' and 'TotalCharges' clearly show that non-churned customers tend to have higher values, while 'MonthlyCharges' indicates a more complex relationship.

3. Data Preprocessing

1. Duplicate value check and treatment

We identified four duplicate records within the dataset and removed them to ensure data integrity.

2. Data Cleaning

- **Target Variable:** The Churn column initially contained inconsistent strings (e.g., ' No ', 'NO', 'YES', 'yes', 'Yes', 'No', 'no', ' Yes '). These were standardized and converted into ‘yes’ and ‘no’.
- **Financial Feature Correction:** MonthlyCharges and TotalCharges were imported as text due to currency symbols ('$' and '£'). We removed these symbols and converted the columns to float types.
- **Payment Method:** Leading and trailing spaces were stripped from categorical features like PaymentMethod to ensure unique categories were not duplicated by formatting errors.

3. Anomalous value check and treatment

- **Negative Value Treatment:** We identified impossible negative values in tenure and TotalCharges. These were treated as data entry errors and replaced with the median value of their respective columns to maintain data distribution without introducing bias.
- **Financial Feature:** Almost all rows (11,871) use dollars, while only a few (183) use pounds. This is likely a typing error, so we changed everything to dollars to keep the data consistent.

4. Missing value check and treatment

- **Missing Value Imputation:** We found 301 missing values in MonthlyCharges and 1,205 in TotalCharges. We filled these values using the median value because it is more reliable than the average when dealing with outliers.

5. Outlier check and treatment

- After checking all numerical columns, only TotalCharges showed outliers. The data is right-skewed, meaning several customers have much higher charges than the rest. However, we decided to keep these points because they represent long-term, high-value customers. Since these are real, valid records, removing them would lose important information.

6. Feature engineering

- **Categorical Encoding:** Since machine learning models require numerical input, categorical variables were converted using One-Hot Encoding.

7. Feature scaling

- To ensure features like TotalCharges (values in thousands) do not mathematically overpower SeniorCitizen (0 or 1), we applied feature scaling (e.g., StandardScalar) to bring all numeric features to a comparable range.

8. Data preparation for modeling

We completed the data preprocessing by scaling the features and splitting the dataset into an 80/20 train-test split. This ensures the model is validated on unseen data to prevent overfitting. The resulting sets are:

- Training Set: 9,640 samples
- Testing Set: 2,411 samples

9. Data leakage handling

To prevent data leakage, we only trained the model on the training set. We kept the testing set completely separate, using it only at the very end to check how the model performs on new, unseen data.

4. Model Building - Baseline Model

1. Metric of Choice

The target variable is binary (churn or no churn) and the dataset is imbalanced (fewer churned customers than non-churned), specific evaluation metrics are more appropriate than simple accuracy.

Here are the chosen metrics with their rationale:

1. **Recall for the 'Churn' class :**

- Rationale: Recall measures the proportion of actual churners that were correctly identified by the model. (minimize False Negatives) Missing a churner means losing a customer, which directly impacts revenue.

2. **Precision for the 'Churn' class:**

- Rationale: Precision measures the proportion of customers predicted as churners who actually churned.(minimize False Positives). High precision ensures that retention efforts are targeted effectively and resources are not wasted on customers who wouldn't churn anyway.

3. **F1-Score:**

- Rationale: The F1-Score is the harmonic mean of Precision and Recall. A high F1-score indicates a good balance between identifying positive cases and not making too many false positive predictions.

4. **ROC AUC (Receiver Operating Characteristic - Area Under the Curve):**

- Rationale: ROC AUC measures the ability of the model to distinguish between positive and negative classes across various classification thresholds. A higher AUC indicates a better model performance in separating churners from non-churners.

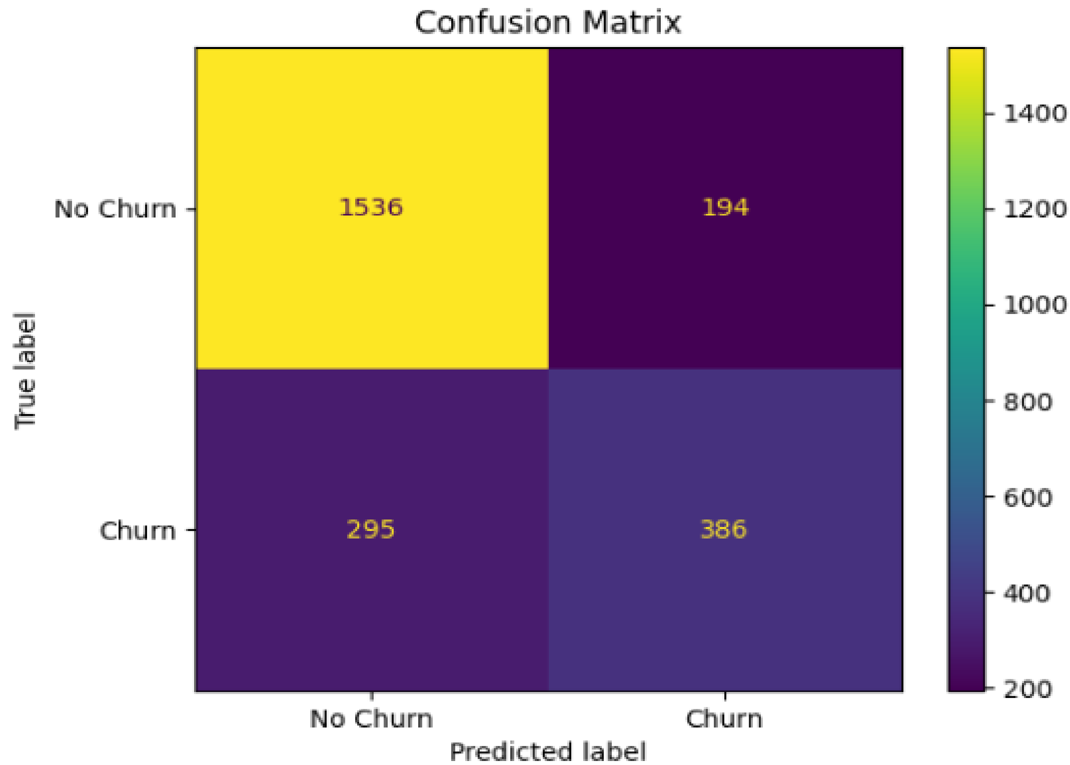
In summary, while we will monitor accuracy, **the primary focus will be on Recall, Precision, and F1-Score for the 'Churn' class**, and **ROC AUC for overall model performance**, to ensure that the model is effective in identifying and acting upon potential customer churn.

In a churn scenario, the cost of **False Negatives** (failing to identify a customer who is about to leave) is much higher than the cost of **False Positives** (offering a discount to someone who was going to stay anyway). The main choice will be **Recall**.

2. Build a baseline model

1. Logistic Regression

Confusion Matrix:



Classification Report:

	precision	recall	f1-score	support
No Churn	0.84	0.89	0.86	1730
Churn	0.67	0.57	0.61	681
accuracy			0.80	2411
macro avg	0.75	0.73	0.74	2411
weighted avg	0.79	0.80	0.79	2411

ROC AUC Score: 0.8464

Fig 18: Logistic Regression

2. Decision Tree Classifier

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Confusion Matrix (Decision Tree):
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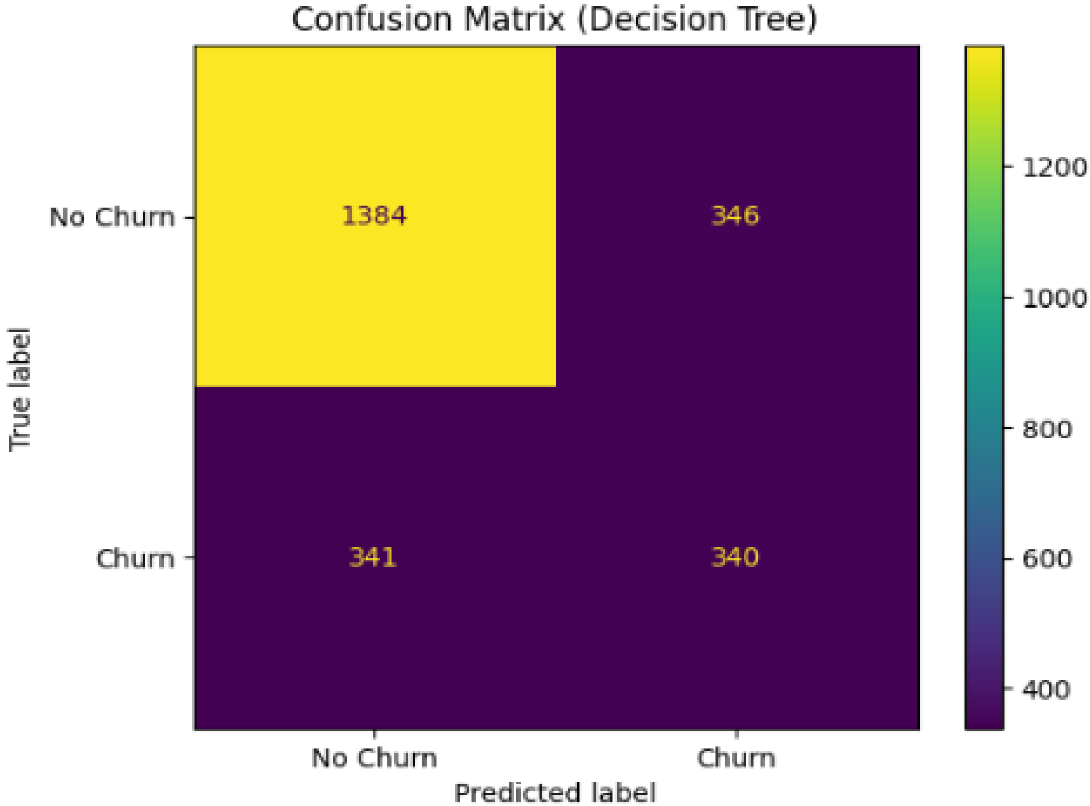


```
Classification Report (Decision Tree):
              precision    recall  f1-score   support

    No Churn       0.80      0.80      0.80      1730
       Churn       0.50      0.50      0.50       681

    accuracy                           0.72      2411
   macro avg       0.65      0.65      0.65      2411
weighted avg       0.72      0.72      0.72      2411


ROC AUC Score (Decision Tree): 0.6493
```

Fig 19: Decision Tree Classifier

3. K-Nearest Neighbors

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Confusion Matrix (K-Nearest Neighbors):
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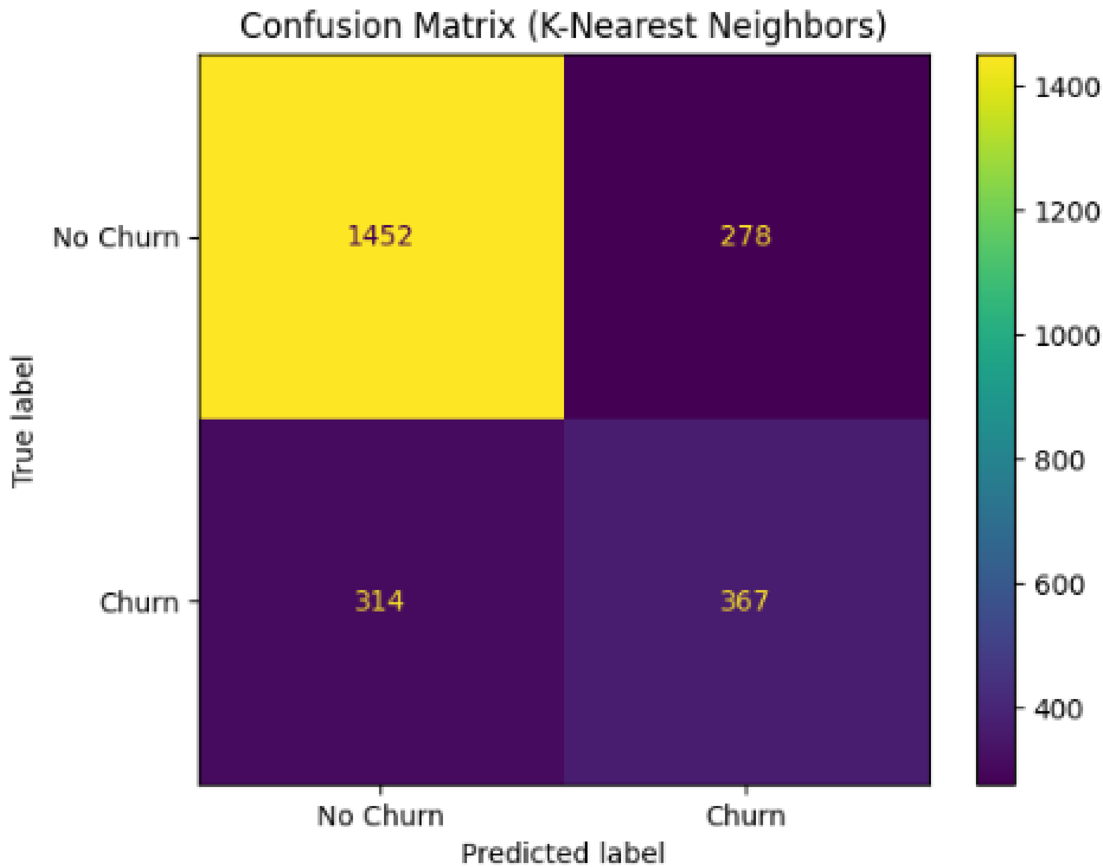


```
Classification Report (K-Nearest Neighbors):
              precision    recall  f1-score   support

    No Churn       0.82      0.84      0.83      1730
       Churn       0.57      0.54      0.55       681

    accuracy                           0.75      2411
   macro avg       0.70      0.69      0.69      2411
weighted avg       0.75      0.75      0.75      2411


ROC AUC Score (K-Nearest Neighbors): 0.7843
```

Fig 20: K-Nearest Neighbors

Comparing all three baseline models

- **Logistic Regression:**
  - Precision (Churn): 0.67
  - Recall (Churn): 0.57
  - F1-Score (Churn): 0.61
  - ROC AUC Score: 0.8464
- **Decision Tree Classifier:**
  - Precision (Churn): 0.50
  - Recall (Churn): 0.50
  - F1-Score (Churn): 0.50
  - ROC AUC Score: 0.6493
- **K-Nearest Neighbors Classifier:**
  - Precision (Churn): 0.57
  - Recall (Churn): 0.54
  - F1-Score (Churn): 0.55
  - ROC AUC Score: 0.7843
- **Conclusion:**

Among the three baseline models, **the Logistic Regression model performed the best**. It achieved the highest scores across all the key evaluation metrics.

Thank You.